Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your answer. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A system absorbs **70 J** heat and does **30 J** work. **Find its internal-energy change** using $\Delta U=Q-W_{by}$.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: v = f λ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength **3 m** and frequency **2 Hz** travels at v=f λ=**6 m/s**. At the same speed, frequency **3 Hz** would require wavelength **2 m**.

Your turn: Draw both waves over the same **6 m** length and **label their frequencies**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two waves travel in the same ideal medium at the same speed. One has twice the frequency. **Compare wavelengths**.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

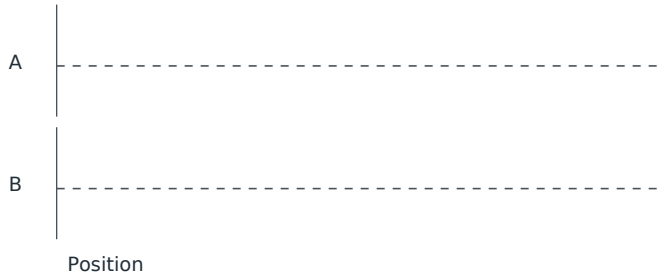


Marzano check - circle: ① ② ③

04 Apply the physics

A wave simulator shows frequency **4 Hz** and wavelength **2 m**. **Calculate wave speed**. Set the frequency to **8 Hz** while keeping speed fixed. **Find the new wavelength** and **sketch both patterns over the** same distance.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student writes v=f/λ. **Check units** and **explain why** that is not a speed.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Sunlight carries energy across **4** million km of nearly empty space in a simplified example. **Name the transfer mechanism** and **explain why matter** is unnecessary. **Label temperatures**, samples and relevant energy transfers.

Sentence starter: I think … . I can point to …, which shows …, so … .

Sample A

Sample B

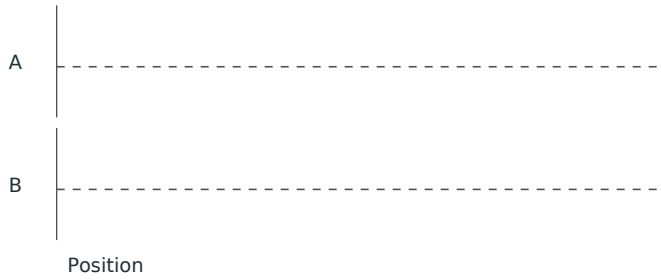
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

f=**3 Hz**,λ=**5 m**. **Find v**.

Sentence starter: I found … [unit]. This value means … in this situation.

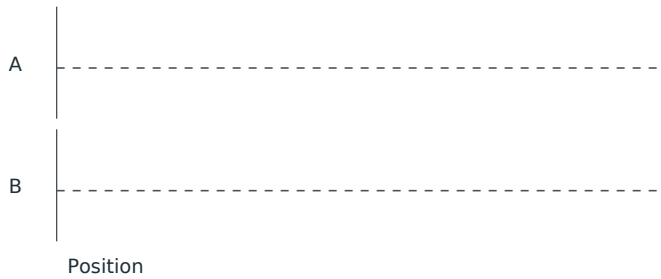


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Create three frequency-wavelength pairs that all give **12 m/s**.
2. **Sketch one pair**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: 1 2 3